Linear Algebra for Data Science, Lecture 3

Singular Value Decomposition

Example lecturer

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Outline

1. Existence of the SVD
2. Low-rank approximation
3. Numerical considerations

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Definition (SVD)

Every A in R^{m x n} can be written as A = U S V^T with

U in R^{m x m}, V in R^{n x n} orthogonal and S diagonal with

s_1 >= s_2 >= ... >= s_r > 0, r = rank(A).

The columns of U (V) are the left (right) singular vectors.

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Why does the SVD always exist?

A^T A is symmetric positive semidefinite => spectral theorem applies:

A^T A = V L V^T with L = diag(l_1,...,l_n), l_i >= 0.

Set s_i = sqrt(l_i) and u_i = A v_i / s_i for s_i > 0; complete U orthonormally.

Contrast: the eigendecomposition needs A square and diagonalisable.

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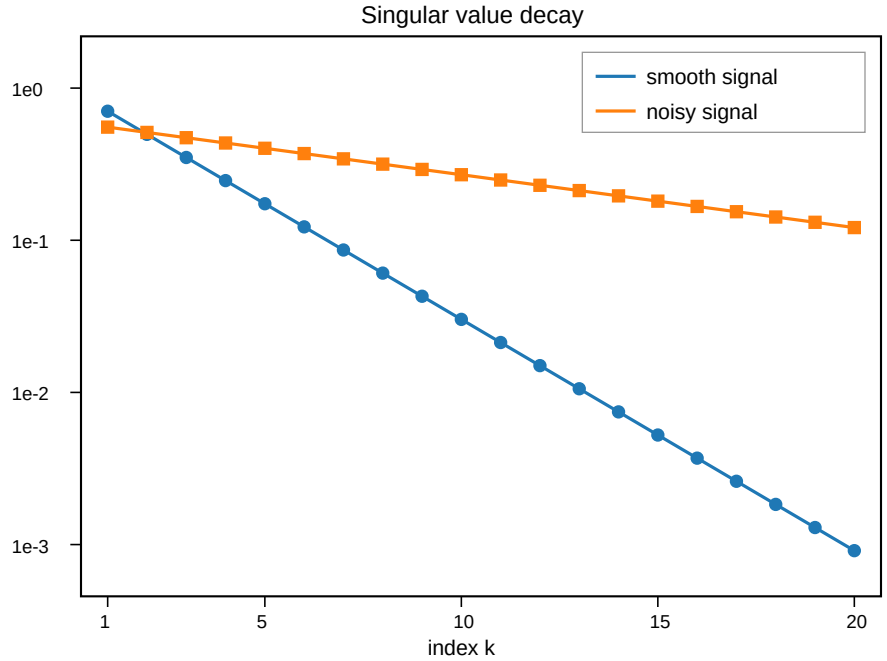
Singular value decay in practice

Smooth signals: exponential decay; noise: flat tail.



Effective rank = number of sigma_k above the noise floor

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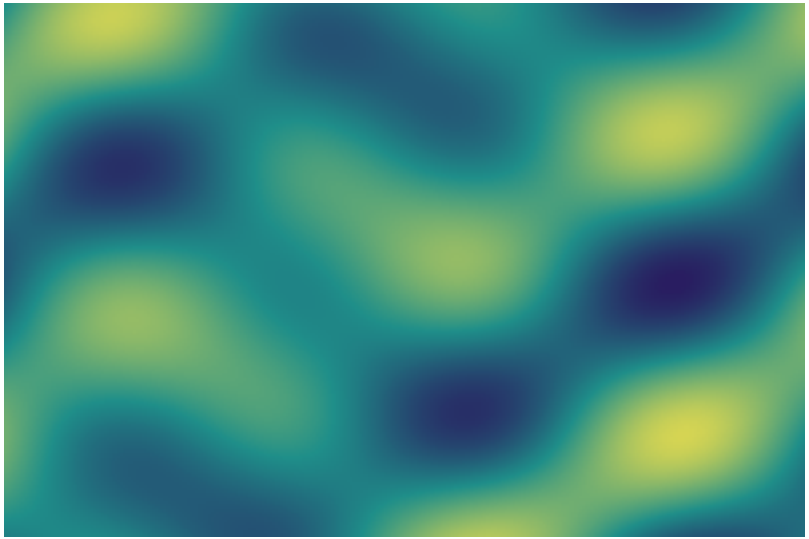
The data matrix

Rows = samples, columns = features.



Centering: subtract the column means

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Eckart-Young-Mirsky theorem

For k < r, the best rank-k approximation in Frobenius and spectral norm is

$A_k = \sum_{i<=k} s_i u_i v_i^T$, with

$||A - A_k||_2 = s_{k+1}$, $||A - A_k||_F = \sqrt{\sum_{i>k} s_i^2}$.

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Comparison of decompositions

	Eigen	SVD
Matrix shape	square	any
Always exists	no	yes
Basis	eigenvectors (not nec. orthogonal)	two orthonormal bases
Cost (n x n)	O(n^3)	O(n^3)

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Numerical considerations

Never form A^T A explicitly to compute the SVD: squares the condition number.

kappa(A^T A) = kappa(A)^2 => small singular values are lost in floating point.

Use Golub-Kahan bidiagonalisation (LAPACK gesvd / gesdd).

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Summary

SVD exists for every matrix; Eckart-Young gives optimal low-rank approximation;

avoid A^T A numerically.

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